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DATA ANALYSIS WITH PYTHON

CLUSTERING ANALYSIS REPORT
Household Segmentation for Grocery Retail

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Abstract

This report investigates whether the 2,500 loyalty-card households of a mid-sized U.S. grocery retailer can be meaningfully segmented based on two years of purchasing behavior. Using a 15-feature behavioral vector derived from transaction records, we apply and compare three unsupervised learning algorithms—K-Means, Agglomerative Hierarchical Clustering (Ward), and DBSCAN. Cluster stability is assessed via Adjusted Rand Index across algorithm pairs. The resulting segments are profiled through radar charts and demographic cross-tabulation, culminating in differentiated marketing recommendations. We identify K=2 distinct household segments with moderate inter-algorithm agreement (ARI=0.552 between K-Means and Hierarchical). The most discriminating features are `campaign_exposure_count`, `active_weeks_ratio`, and monetary value.

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1 Introduction

1.1 Business Context

Modern grocery retail operates on thin margins where customer intelligence is a primary competitive advantage. Retailers who understand the behavioral heterogeneity of their customer base can allocate marketing spend more efficiently, design targeted promotions, and improve retention for high-value segments. Despite access to rich transactional data through loyalty programs, many retailers continue to treat their customer base uniformly, running identical campaigns across all households regardless of behavioral differences. This study uses the Dunnhumby 'Complete Journey' dataset — a two-year record of transactions from 2,500 households across 582 stores and 30 marketing campaigns — to investigate whether meaningful behavioral segments exist and what they imply for retail strategy.

1.2 Research Question

The central question driving this analysis is: Do natural customer groups exist within the household population with sufficiently distinct behavioral profiles to justify differentiated marketing strategies? This question is important because if such groups do not exist or if they exist but do not differ in actionable ways then segmentation provides no value over mass marketing. Conversely, if robust, interpretable segments emerge, they provide the foundation for personalized retention, cross-sell, and promotion strategies.

1.3 Methodology Overview

We follow a six-step pipeline from raw transaction data to business recommendations:

- Data Integration & Cleaning — consolidating 5 source files into analysis-ready tables.
- Exploratory Data Analysis — validating that behavioral variance justifies clustering.
- Feature Engineering — constructing 15 household-level behavioral indicators across 6 MECE groups.
- Clustering — comparing K-Means, Hierarchical, and DBSCAN on the same normalized feature matrix.
- Evaluation — using Silhouette Score, Elbow Method, Davies-Bouldin Index, and Adjusted Rand Index.
- Profiling & Recommendations — describing segments and proposing differentiated strategies.

2 Dataset Description

2.1 Overview

The Dunnhumby 'Complete Journey' dataset is a longitudinal record of purchasing behavior from a mid-sized U.S. grocery chain. It captures the complete shopping history of 2,500 households enrolled in the retailer's loyalty program over approximately two years (104 weeks). The dataset is structured across eight relational tables covering transactions, product metadata, household demographics, and marketing campaign exposure. Key statistics: 2,500 households, 275,000+ shopping baskets, 582 stores, 30 campaigns, basket sizes ranging from 1 to 80 items with a median of approximately 7 items per trip.

2.2 File Schema

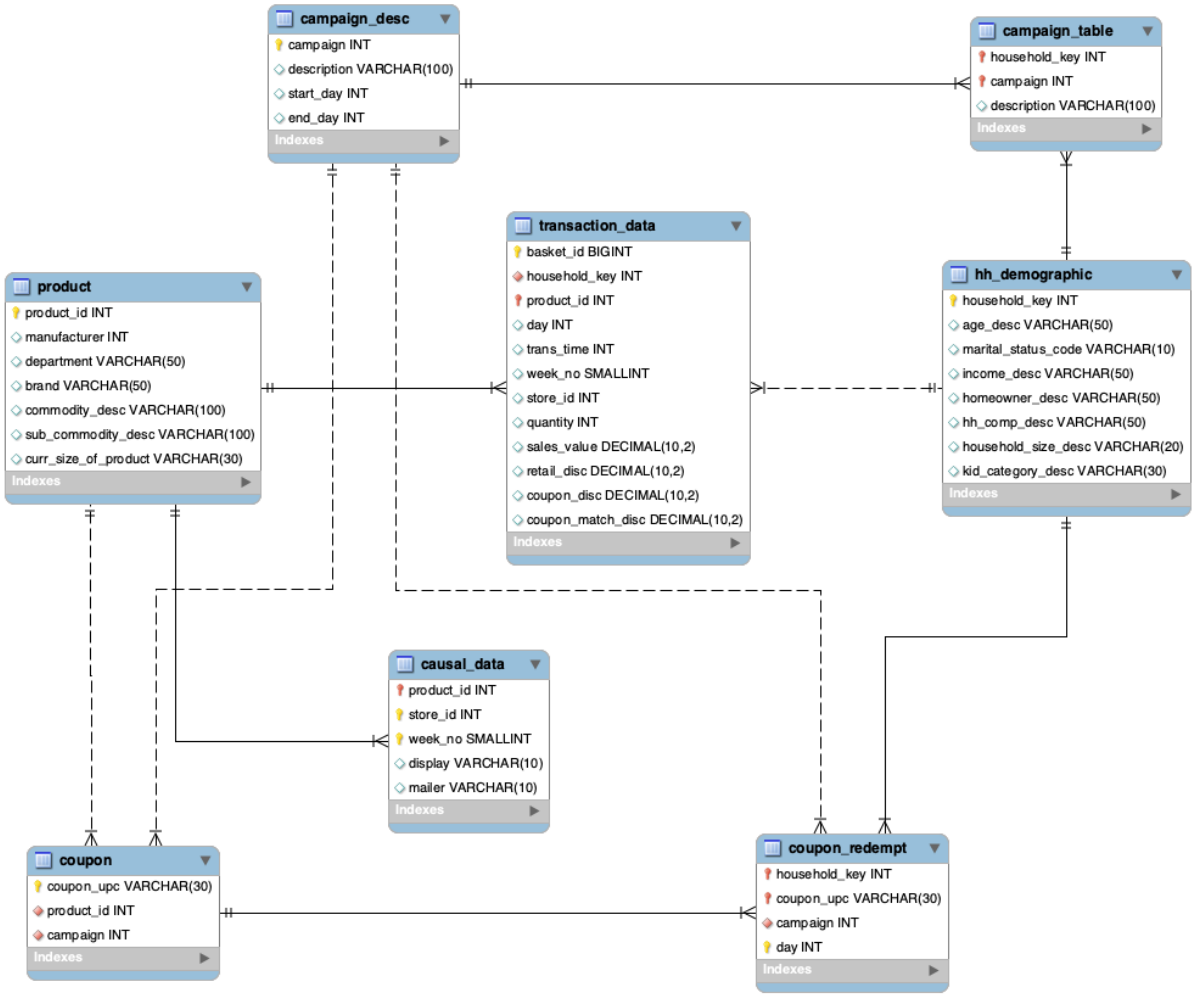

Figure 1: Entity-Relationship Diagram of source files

File	Primary Key(s)	Rows	Role in Analysis
transaction_data	household_key, BASKET_ID	2.6M	Core — all features derived here
product	PRODUCT_ID	∼90K	Enrichment: DEPARTMENT, BRAND
hh_demographic	household_key	∼1.5K	Post-hoc profiling only (∼60% coverage)
campaign_table	household_key, CAMPAIGN	7,208	Campaign exposure per household
campaign_desc	CAMPAIGN	30	Campaign type and date range
coupon_redempt	household_key, COUPON_UPC	2,318	Coupon usage per household
coupon	COUPON_UPC	∼1.2K	Coupon-product mapping
causal_data	PRODUCT_ID, STORE_ID, WEEK_NO	∼4M	Not used (captured via discount_rate)

Table 1: File Schema and Data Usage Summary

2.3 Key Data Characteristics

SALES_VALUE interpretation: SALES_VALUE represents net retailer revenue after loyalty card discount, not the price paid by the customer. We use SALES_VALUE consistently as the household value metric because it measures economic contribution to the retailer, which is the relevant dimension for marketing prioritization.

RETAIL_DISC sign convention: RETAIL_DISC is stored as a negative value (the discount subtracted from gross price). All discount-rate calculations in this analysis apply abs() before computing ratios, ensuring all derived features are non-negative and interpretable.

hh_demographic coverage limitation: Demographic records are available for approximately 32% of the 2,500 households. This partial coverage precludes using demographic variables as clustering inputs — imputing categorical fields such as income brackets would introduce systematic bias. Demographic data is used exclusively for post-hoc cluster profile enrichment.

3 Data Cleaning

Cleaning decisions follow one principle: preserve analytical integrity without discarding valid behavioral signal. Every decision is documented with a justification. Before cleaning, the transaction data contained 2,595,732 rows. After removing 18,850 rows (0.73%) with zero or negative sales values (returns/voids), the final cleaned dataset contains 2,576,882 rows across 2,500 unique households. There were 0 duplicate rows identified.

Issue	**Decision**	**Rationale**
Duplicate (household, basket, product) rows	Remove exact duplicates	Same product recorded twice in one basket is a data entry artifact, not a genuine purchase event
`SALES_VALUE` ≤ 0	Remove rows	Zero or negative values represent returns and voids — not purchasing behavior relevant to segmentation
`DEPARTMENT` null after join	Retain rows, flag as 'Unknown'	Product metadata gaps should not bias total household spend figures; 'Unknown' becomes a valid department bucket
`hh_demographic` nulls (∼40%)	Do not impute; post-hoc profiling only	Categorical income and marital status fields cannot be meaningfully imputed; doing so would introduce bias into cluster validation
Households with < 4 active weeks	Set `spending_trend_slope` $= 0$	Fewer than 4 weekly data points yield an OLS slope with no statistical reliability; zero indicates no detectable trend
Coupon/campaign non-participants	Fill 0	Absence of redemption or campaign exposure is a genuine behavioral signal (non-user), not missing data

Table 2: Data Cleaning Summary

4 Exploratory Data Analysis

4.1 Purpose

Exploratory analysis serves a specific purpose here: to determine whether behavioral variance in the data is sufficient to make clustering meaningful. We examine five dimensions — spending distribution, temporal trends, category diversity, RFM structure, and promotion behavior — and draw an explicit conclusion about clustering viability before proceeding to feature engineering.

4.2 Spending and Frequency Distribution

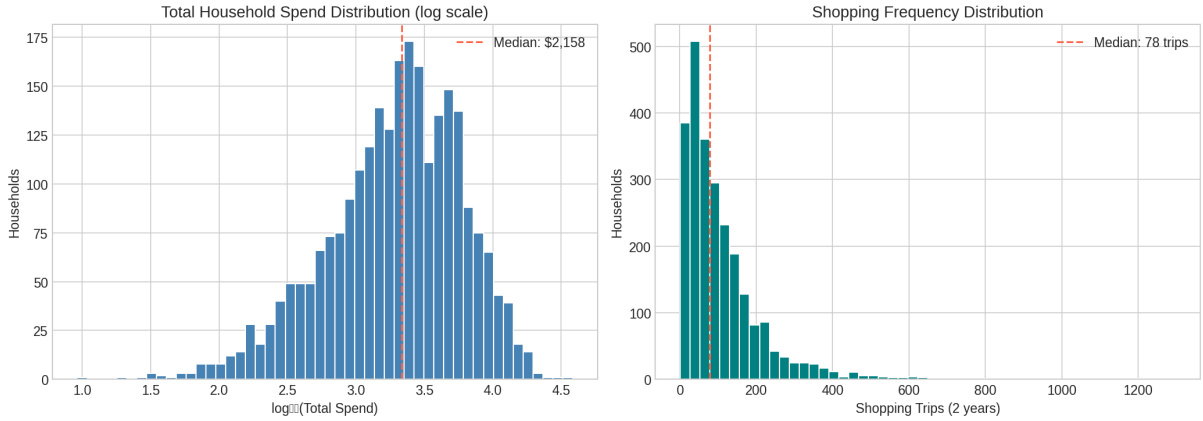

Figure 2: Total household spend (log scale) and shopping trip frequency distributions

Both the total household spend and shopping trip frequency distributions are right-skewed with high variance, confirming that the household population is heterogeneous and uniform treatment would ignore substantial behavioral differences. Total spend ranges from $8 to $38,320 with a median of $2,158. The log-scale distribution is approximately bell-shaped, confirming the underlying raw distribution is log-normal with a long right tail. Shopping frequency ranges from 1 to 1,298 trips with a median of 78. Even in raw scale, frequency is heavily right-skewed — most households shop under 150 times while a small group exceeds 400. Both distributions show high variance, confirming the household population is heterogeneous. Treating all households identically would systematically ignore these differences, so clustering is warranted.

4.3 Temporal Spending Trends

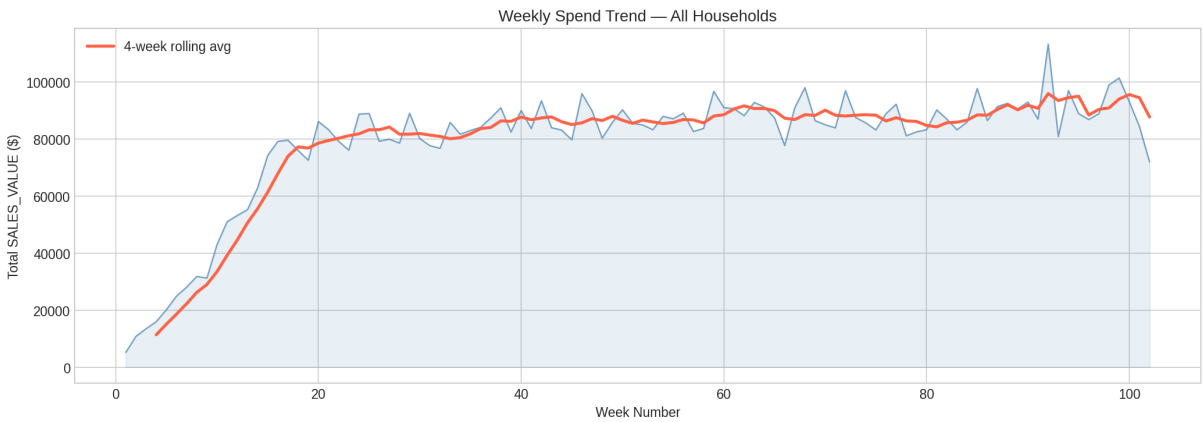

Figure 3: Weekly total SALES_VALUE with 4-week rolling average overlay

At the aggregate level, weekly spend shows a sharp initial increase during the first 20 weeks before stabilizing. Weeks 1–18 show a sharp ramp-up from ∼$5,000 to ∼$80,000,

reflecting loyalty program onboarding rather than genuine behavioral growth. From week 18 onward, aggregate spend plateaus around $80,000–$95,000 with no secular trend. A spike near weeks 90–95 (∼$115,000) likely reflects a seasonal event or concentrated promotion. The key insight is the contrast between this flat aggregate and the wide distribution of individual slopes: 1,137 of 2,500 households carry negative `spending_trend_slope` values, meaning nearly half are in spending decline relative to their own baseline — a churn signal invisible at the aggregate level.

4.4 Category Landscape

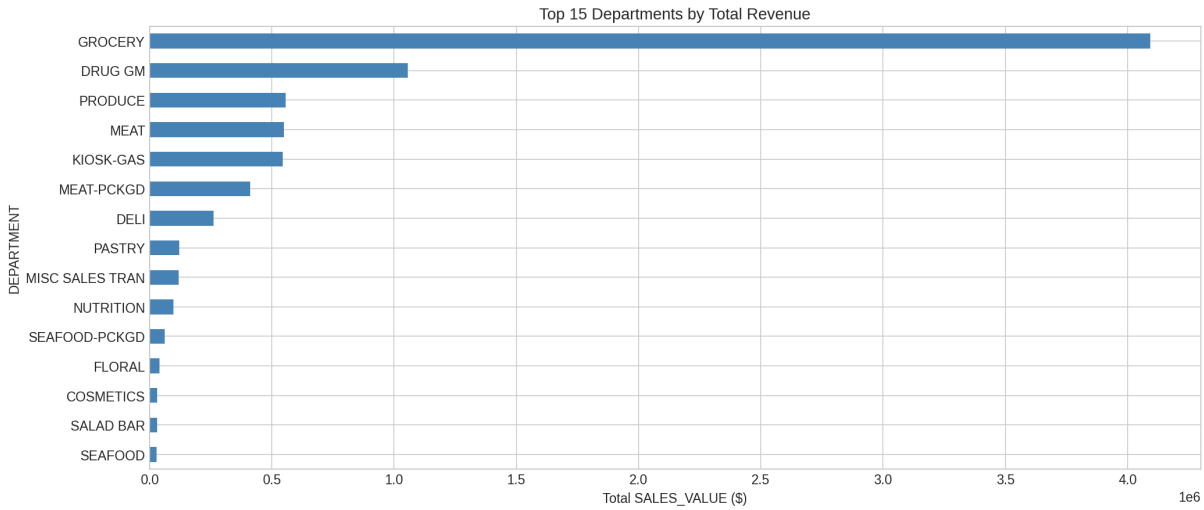

Figure 4: Top 15 departments by total SALES_VALUE

GROCERY dominates at ∼$4M, approximately four times DRUG GM (∼$1M). PRODUCE, MEAT, and KIOSK-GAS form a secondary tier around $0.5M each. The remaining departments fall below $0.4M, forming a long tail. This extreme concentration means all households are GROCERY-heavy at the aggregate level, but what differentiates them is secondary allocation. A household spreading spend across PRODUCE, MEAT, DELI, and NUTRITION has a low HHI (generalist); one concentrating outside GROCERY in a single category has high HHI (specialist). The long tail of available departments makes `category_hhi` a meaningful differentiator.

4.5 RFM Structure

The population forms one continuous diagonal cloud from low-frequency/low-spend to high-frequency/high-spend with no sharp visual breaks. Color aligns cleanly with position: lapsed households (red/yellow) cluster in the bottom-left low-value region, active households (green) dominate the upper-right. A small pocket of red dots at very low frequency represents inactive households who rarely shopped and have since churned. The

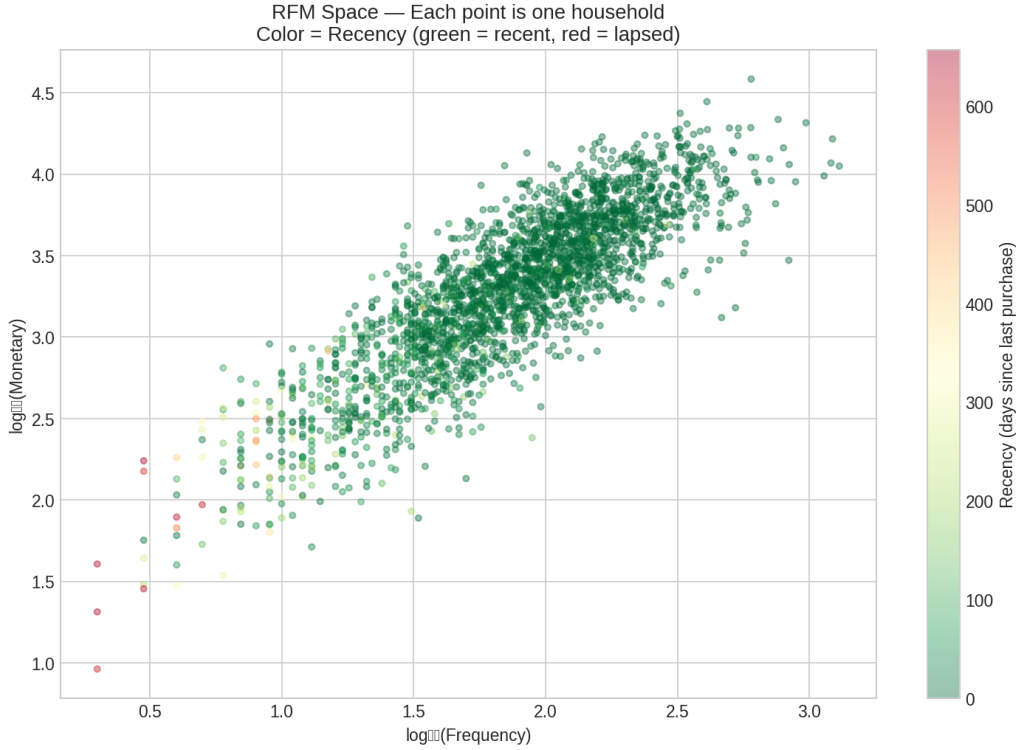

Figure 5: RFM scatter plot: $\log_{10}$(Frequency) vs $\log_{10}$(Monetary), colored by Recency.

absence of discrete visual clusters indicates that segment boundaries will be soft — consistent with the low Silhouette scores later — but the strong value gradient along the diagonal provides a clear structural basis for a two-group split.

4.6 Promotion Behavior

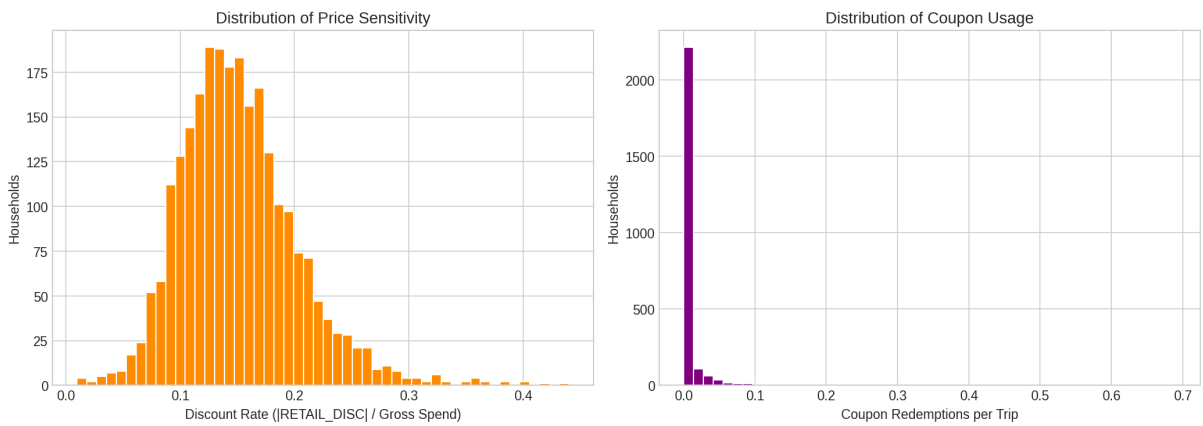

Figure 6: Distribution of discount rate and Coupon redemptions per trip

Discount rate is bell-shaped, centered around 0.13–0.15, ranging 0.01–0.44. Because all loyalty households receive automatic discounts, this reflects program structure as much as individual behavior — its discriminating power between segments is limited. Coupon

redemption is the more informative feature: over 2,200 of 2,500 households show zero redemptions, while a small tail extends to 0.69 redemptions per trip. This zero-inflation means coupon usage is a rare but high-signal indicator that identifies a behaviorally distinct subgroup actively seeking savings beyond the standard loyalty discount.

4.7 EDA Conclusion

Based on the analysis above:

- Spend and frequency distributions show high variance across households — treating all households identically would ignore substantial behavioral differences.
- The RFM scatter shows a continuous gradient rather than visually discrete clusters, but the strong variance along the frequency-monetary diagonal and the clear recency color separation indicate meaningful structure exists for a value-based split.
- Temporal trends exist at the aggregate level, suggesting individual trend slopes will carry signal.
- Promotion sensitivity varies widely, justifying `discount_rate` and `coupon_redemption_rate` as distinct features.
- Conclusion: clustering is warranted. We proceed to construct a 15-feature behavioral vector per household.

5 Feature Engineering

5.1 Design Rationale

Feature engineering transforms 2.6 million transaction rows into a single 15-dimensional behavioral vector per household. Each feature was selected because it captures a distinct aspect of purchasing behavior identified as variable in EDA, or represents a known driver of customer lifetime value in grocery retail. Features are organized into six MECE groups to ensure comprehensive coverage without redundancy.

Two features go beyond the standard RFM framework and constitute the primary analytical contribution of this project:

- `spending_trend_slope` is the OLS slope of a household's weekly spend over the two-year observation window. A positive slope indicates a household increasing its spend over time — a growing asset. A negative slope indicates declining spend — a potential churn risk. This dimension is entirely absent from standard RFM analysis and provides the most actionable forward-looking signal in the feature set.
- `category_hhi` (Herfindahl-Hirschman Index) measures the concentration of a household's spending across product categories: $\sum(\text{dept_share}^2)$. A value of 1 indicates all spend in one category (specialist); values approaching 0 indicate evenly distributed spending (generalist). This captures whether a household is a category specialist or a generalist shopper — a behavioral distinction with direct implications for cross-sell strategy.

5.2 Feature Table

Feature	Group	Formula / Source	Business Meaning
`recency`	RFM	max_DAY − last purchase DAY	Days since last purchase — churn proximity
`frequency`	RFM	COUNT(DISTINCT BASKET_ID)	Shopping trips over 2 years
`monetary`	RFM	SUM(SALES_VALUE)	Total value generated for retailer
`avg_basket_value`	Basket	monetary / frequency	Spend per trip
`avg_basket_size`	Basket	SUM(QUANTITY) / frequency	Items per trip — bulk vs. top-up shopper
`std_basket_value`	Basket	STD(basket totals)	Spend consistency — predictable vs. volatile
`n_unique_departments`	Category	COUNT(DISTINCT DEPARTMENT)	Breadth of category purchase
`category_hhi` $\star$	Category	$\sum(\text{dept_share}^2)$	Concentration index — specialist vs. generalist
`national_brand_ratio`	Category	National items / total items	Brand preference — price vs. quality signal
`discount_rate`	Promotion	$\lvert$`RETAIL_DISC`$\rvert$ / gross spend	Price sensitivity — deal-driven vs. loyal
`coupon_redemption_rate`	Promotion	Redemptions / frequency	Coupon engagement per trip
`campaign_exposure_count`	Promotion	COUNT(DISTINCT CAMPAIGN)	Marketing exposure over 2 years
`active_weeks_ratio`	Temporal	Active weeks / 104	Shopping regularity
`spending_trend_slope` $\star$	Temporal	OLS β_1 of weekly spend $\sim$ week	Trajectory — growing vs. declining value
`primary_store_share`	Store	Max store basket visits / frequency	Store loyalty — anchor vs. multi-store shopper
$\star$ = features beyond standard RFM — analytical contribution of this project			

Table 3: Feature Engineering Dictionary

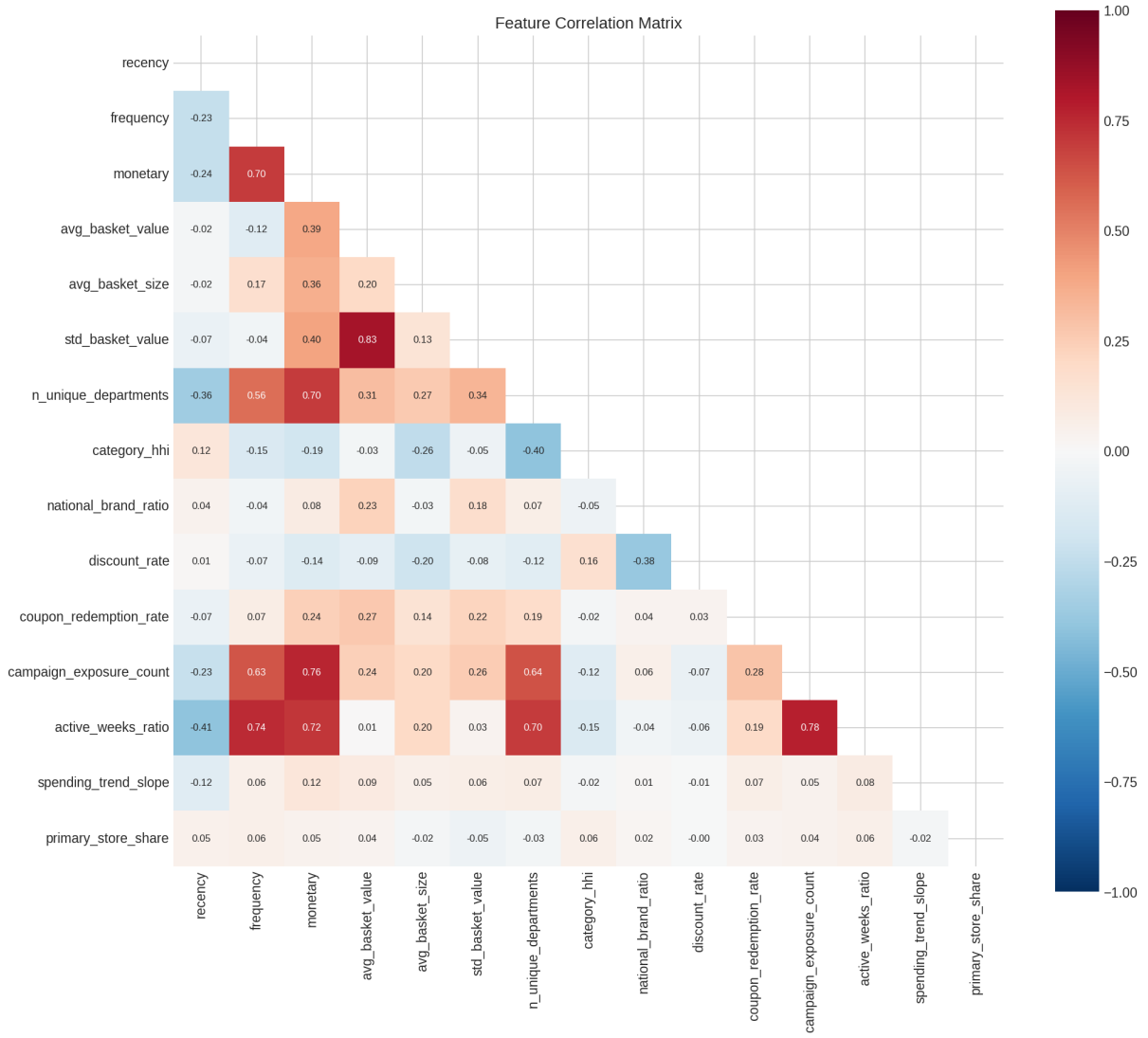

Figure 7: Pearson correlation matrix of all 15 features

5.3 Correlation and Dimensionality Check

No feature pair exceeds $|r| = 0.9$, so all 15 features are retained. The strongest correlations form an engagement-volume cluster: `active_weeks_ratio` ↔ `campaign_exposure_count` (0.78), `active_weeks_ratio` ↔ `frequency` (0.74), `campaign_exposure_count` ↔ `monetary` (0.76). These reflect one underlying dimension — households who shop more also accumulate more campaign exposure and total spend — but no pair is redundant enough to remove. Independently contributing features include `spending_trend_slope`, `primary_store_share`, `coupon_redemption_rate`, and `national_brand_ratio`. The `std_basket_value` ↔ `avg_basket_value` pair at 0.83 is the closest to redundancy but remains within the retention threshold.

8 principal components are needed to explain 80% of variance; 10 components for 90%. The feature space is genuinely multi-dimensional with no single dominant axis, confirming

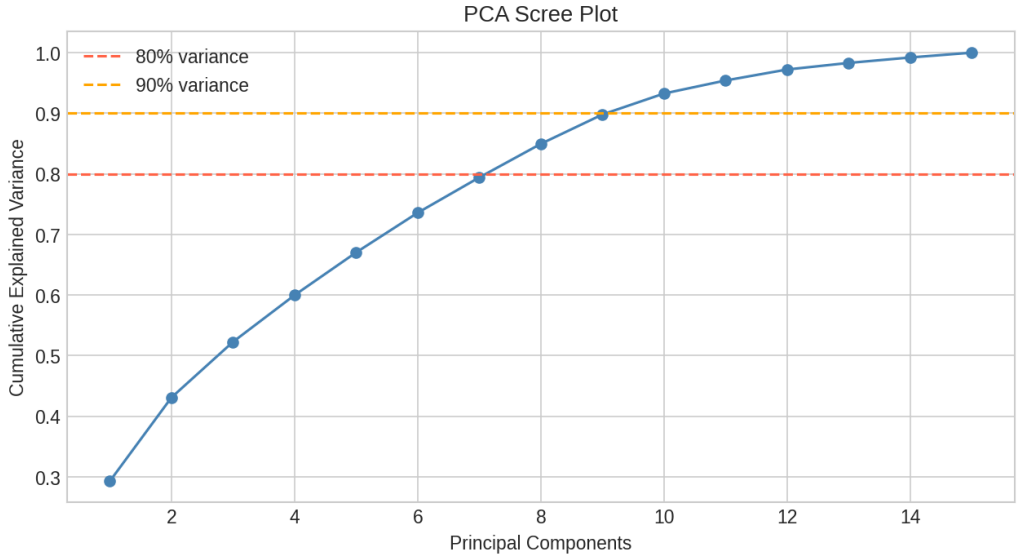

Figure 8: PCA scree plot: cumulative explained variance by number of components

that clustering in the full 15-dimensional standardized space is appropriate rather than on a reduced PCA projection.

6 Clustering Methodology

6.1 Algorithm Selection

K-Means: K-Means partitions n observations into K clusters by iteratively assigning each point to its nearest centroid and updating centroids until convergence. Its primary strength is the interpretability of cluster centroids — each centroid is the mean household in that cluster, providing a natural archetype for profile construction. The assumption of spherical, equal-variance clusters may not hold in all real behavioral data, which is why we cross-validate with Hierarchical clustering. K must be specified in advance, determined here through the Elbow Method and Silhouette Score.

Agglomerative Hierarchical Clustering: Hierarchical clustering builds a dendrogram by iteratively merging the two closest clusters using Ward linkage, which minimizes within-cluster variance at each merge step. Its key advantage is that K does not need to be pre-specified — the dendrogram reveals the natural hierarchy of groupings and allows visual inspection of where meaningful cuts exist. We use the same K as K-Means to enable direct comparison via ARI.

DBSCAN: DBSCAN identifies clusters as dense regions separated by lower-density regions. It does not require K to be pre-specified and can find arbitrarily-shaped clusters. Crucially, it labels low-density points as noise (-1) — households that do not belong to any cluster. These outlier households may represent unusual spending patterns worth investigating separately. DBSCAN requires two hyperparameters: `eps` (neighborhood radius) and `min_samples` (minimum points to form a core). We select `eps` from the elbow of the k-distance plot and set `min_samples` = n_features + 1 per the original algorithm recommendation.

6.2 K Selection for K-Means and Hierarchical

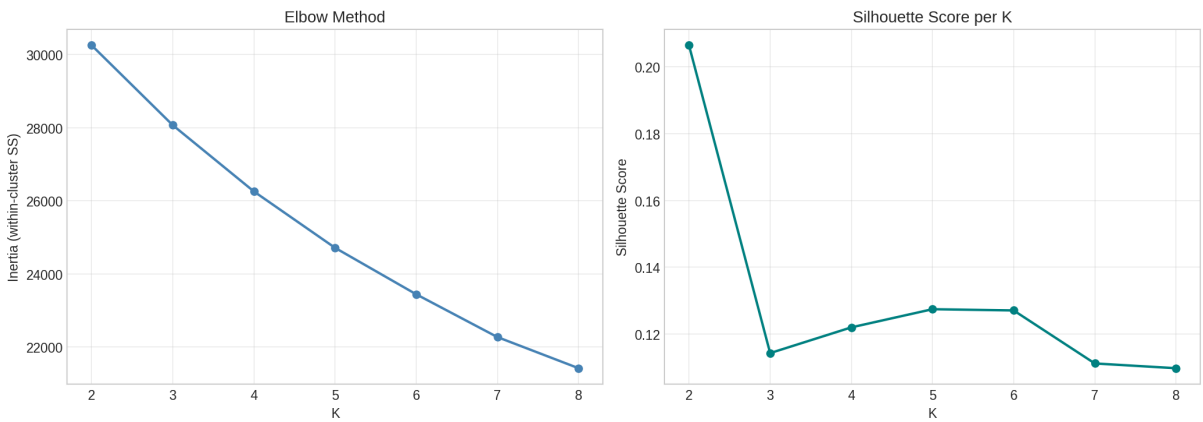

Figure 9: Inertia vs. K (Elbow Method) and Silhouette Score vs. K

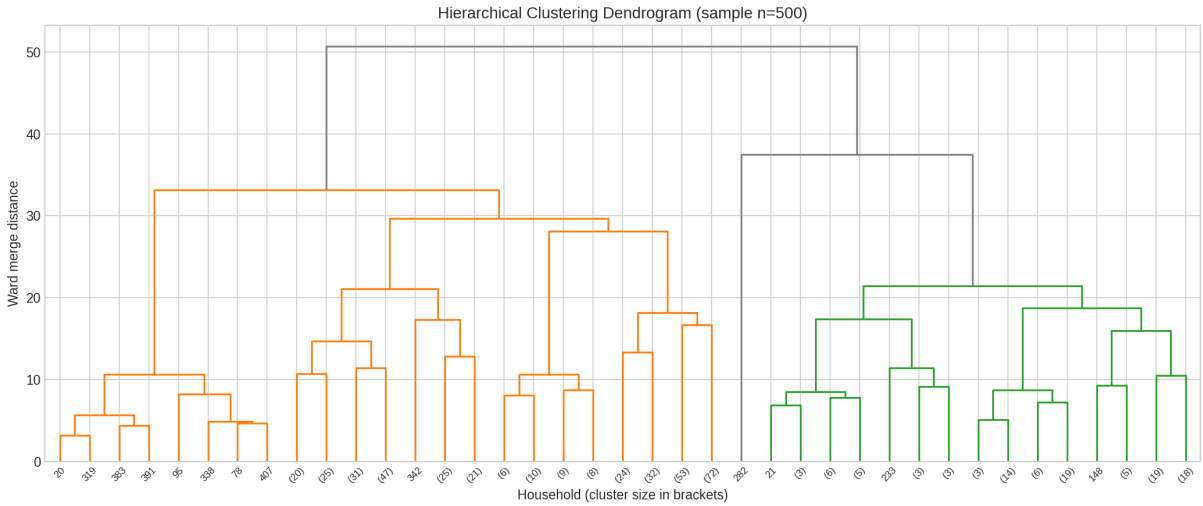

Figure 10: Ward linkage dendrogram on a 500-household sample

The Elbow plot shows inertia declining continuously from K=2 to K=8 with no sharp inflection — no single K stands out as geometrically optimal. The Silhouette plot provides a clear answer: score peaks at K=2 (0.206) and drops sharply to 0.114 at K=3, with only modest recovery thereafter. The dendrogram corroborates this: the only unambiguous cut is at Ward distance ∼50, producing two branches. The left branch merges at lower internal distance (∼33), indicating a more homogeneous group; the right branch is more internally dispersed. Both methods agree on K=2. Although the absolute Silhouette value of 0.206 is modest — reflecting the soft, overlapping boundaries visible in the RFM scatter — K=2 is the statistically supported choice and produces the most business-interpretable split: an engaged high-value segment versus a low-engagement low-value segment.

K	Silhouette ↑	Davies-Bouldin ↓	Calinski-Harabasz ↑
2	0.206	1.826	599
3	0.114	2.118	420
4	0.122	1.960	356
5	0.127	1.835	323
6	0.127	1.733	299
7	0.111	1.810	284
8	0.110	1.816	267

Table 4: K selection metrics for K = 2..8

7 Evaluation Results

7.1 Algorithm Performance

K-Means outperforms Hierarchical on all three geometric metrics at K=2: higher Silhouette (0.206 vs 0.159), lower DBI (1.826 vs 2.062), and higher CH (599 vs 487). More importantly, K-Means cluster centroids are directly interpretable as the mean household in each segment — a natural archetype for profile construction. For these reasons, K-Means labels are used as the primary segmentation in all subsequent profiling and business recommendation sections.

Algorithm	**K**	**Silhouette ↑**	**Davies-Bouldin ↓**	**Calinski-Harabasz ↑**
K-Means	2	0.206	1.826	599
Hierarchical	2	0.159	2.062	487
DBSCAN	1	—	—	N/A

Table 5: Algorithm Comparison

As a sanity check, random label assignment yields Silhouette ≈ 0.000, confirming that the K-Means structure is non-trivial and not an artifact of the algorithm. Clustering on PCA-reduced features (8 components, retaining 80% variance) improves separation to Silhouette $= 0.236$ and DBI $= 1.641$, compared to 0.206 and 1.826 on the full 15-dimensional space. The Adjusted Rand Index between the two assignment vectors is 0.992, indicating that dimensionality reduction recovers the same underlying structure while removing residual noise — cluster membership is effectively unchanged and all profiling results remain valid.

7.2 Cluster Stability — Adjusted Rand Index

The Adjusted Rand Index (ARI) measures agreement between two clustering solutions, corrected for chance. ARI $= 1$ indicates perfect agreement; ARI $= 0$ indicates agreement no better than random. We compute pairwise ARI to assess whether the cluster structure is robust to algorithmic choice. High ARI between K-Means and Hierarchical (> 0.75) indicates the structure exists in the data, not just in one method's assumptions.

ARI of 0.552 between K-Means and Hierarchical indicates the two-cluster structure is partially robust across methods. It falls short of the 0.75 strong-agreement threshold, reflecting the soft boundaries visible throughout the analysis. However, both algorithms independently place the same core households in each group — the disagreement is concentrated at the boundary zone between clusters, not in their cores. The cluster structure is real but not sharply defined.

Algorithm Pair	ARI	Interpretation
K-Means ↔ Hierarchical	0.552	Moderate agreement — both methods recover the same broad structure, though not perfectly aligned at the boundary
K-Means ↔ DBSCAN	0.003	No meaningful agreement — DBSCAN finds one large cluster and 25 noise points, incompatible with K-Means partition
Hierarchical ↔ DBSCAN	0.001	Same reason as above

Table 6: Adjusted Rand Index Comparisons

7.3 DBSCAN Noise Analysis

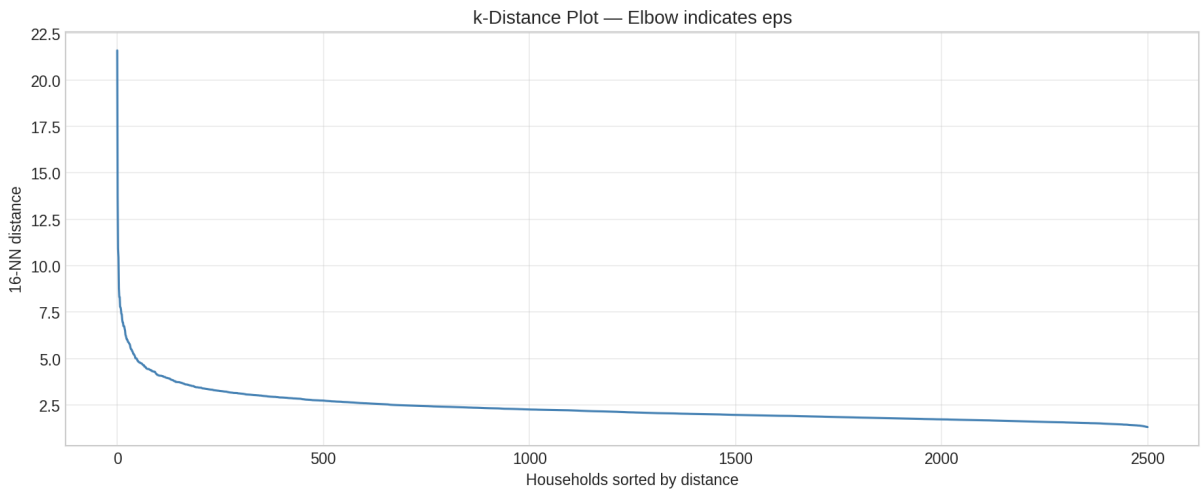

Figure 11: k-distance plot used to select `eps` for DBSCAN

The curve drops sharply from ∼22 at the first few households to ∼5 by household 50, then transitions into a gradual, near-linear decline to ∼1.5 at household 2,500 with no clear elbow. The only sharp inflection (elbow) occurs around distance 4.5 to 5.0, which dictates our eps choice. At eps=4.5 and min_samples=16, DBSCAN assigns 2,475 households to a single cluster and flags 25 (1.0%) as noise. The 25 noise households represent genuine behavioral outliers — extreme values on engagement or spending dimensions — rather than a distinct behavioral group. Their distribution across K-Means clusters is diffuse, confirming they are isolated outliers rather than a coherent segment missed by the other algorithms. The degenerate result (1 cluster) suggests the household space lacks clear density boundaries, consistent with the soft silhouette scores — confirming DBSCAN is not suited for this dataset.

8 Cluster Profiles

8.1 Cluster Visualization

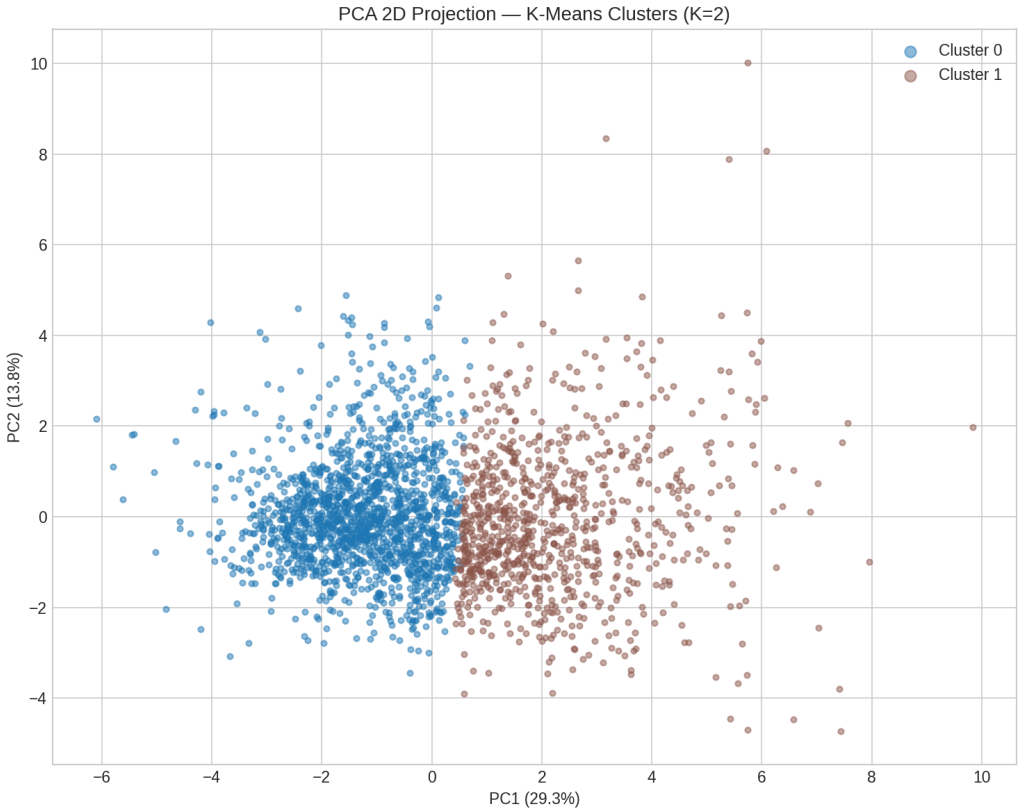

Figure 12: PCA 2D projection of all 2,500 households colored by K-Means cluster

Figure 12 (PCA) projects all households onto the first two principal components, which together explain 43.1% of variance. Cluster 1 occupies the positive PC1 region while Cluster 0 occupies the negative-to-zero region, with substantial overlap in the 0–2 range of PC1. PC2 provides no additional separation, confirming the split is driven by the first principal component — the engagement/value axis — and that boundaries are soft. Figure 13 (t-SNE) reveals slightly cleaner separation than PCA: Cluster 1 concentrates in the right half of t-SNE space while Cluster 0 fills the left and center, with a smaller overlap zone than in the PCA projection. This indicates some non-linear structure that PCA compresses. Cluster 1 appears more spatially compact in t-SNE space — consistent with its higher internal homogeneity observed in the dendrogram. No sub-clusters are visible within either group, confirming K=2 captures the primary structure without meaningful finer granularity.

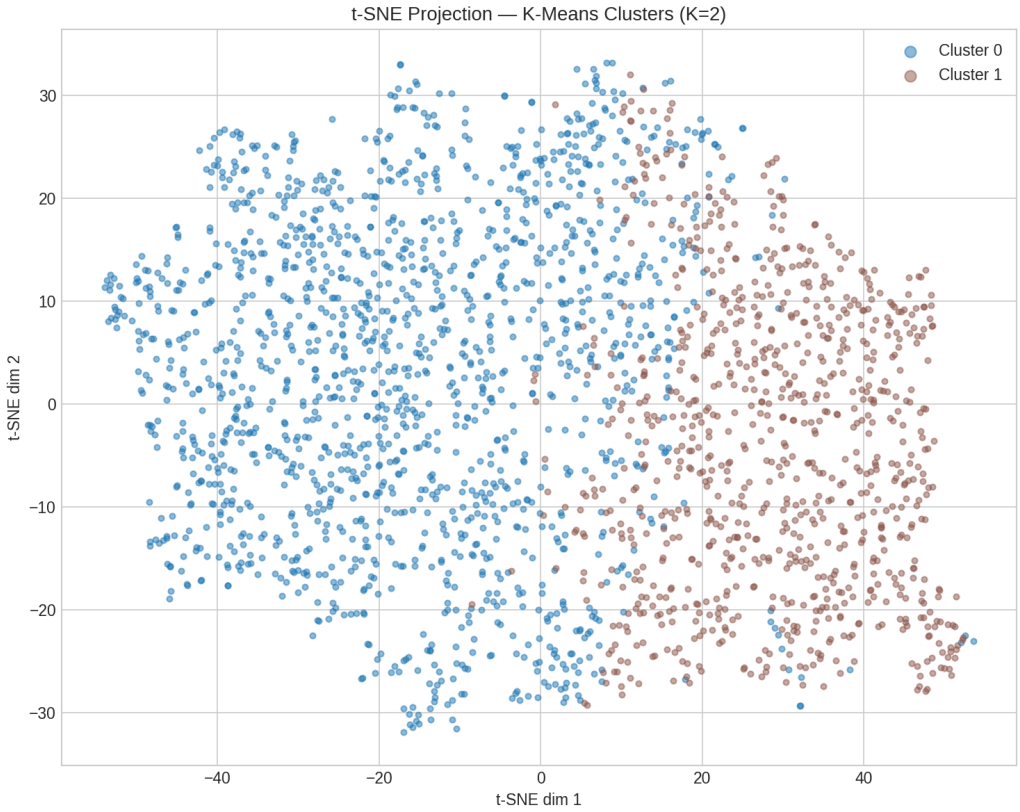

Figure 13: t-SNE 2D projection colored by K-Means cluster

8.2 Cluster Profiles

Cluster 0 — "Passive Occasionals": These households account for 64% of the base but generate only $1,406 on average in total spend — less than a quarter of Cluster 1's value. They shop infrequently (59 trips over two years, active in only 35% of weeks), have low campaign exposure (1.02 campaigns), and a near-zero spending trend slope (−0.003) indicating flat-to-declining engagement. Recency of 36 days suggests many are still nominally active but with low commitment. Their radar profile falls below the population mean across nearly all value dimensions. They are low-engagement, low-investment households whose behavior shows no organic growth.

Cluster 1 — "Committed Regulars": These households represent the retailer's core revenue base. With average monetary value of $6,459 — 4.6× that of Cluster 0 — and 202 shopping trips over two years, they shop roughly every 3–4 days. They are active in 73% of all weeks, have been exposed to an average of 6.2 campaigns, and carry a positive spending trend slope (0.098), indicating growing engagement over the observation window. Recency of just 7 days confirms they are currently active. They shop across more departments (15.5 vs. 10.5) and use coupons at a meaningfully higher rate (0.016 vs. 0.001). Their radar profile consistently exceeds the population mean across all engagement and value dimensions. Despite higher absolute spend, their discount rate (0.143) is slightly lower than Cluster 0 (0.156), suggesting loyalty rather than deal-seeking drives their behavior.

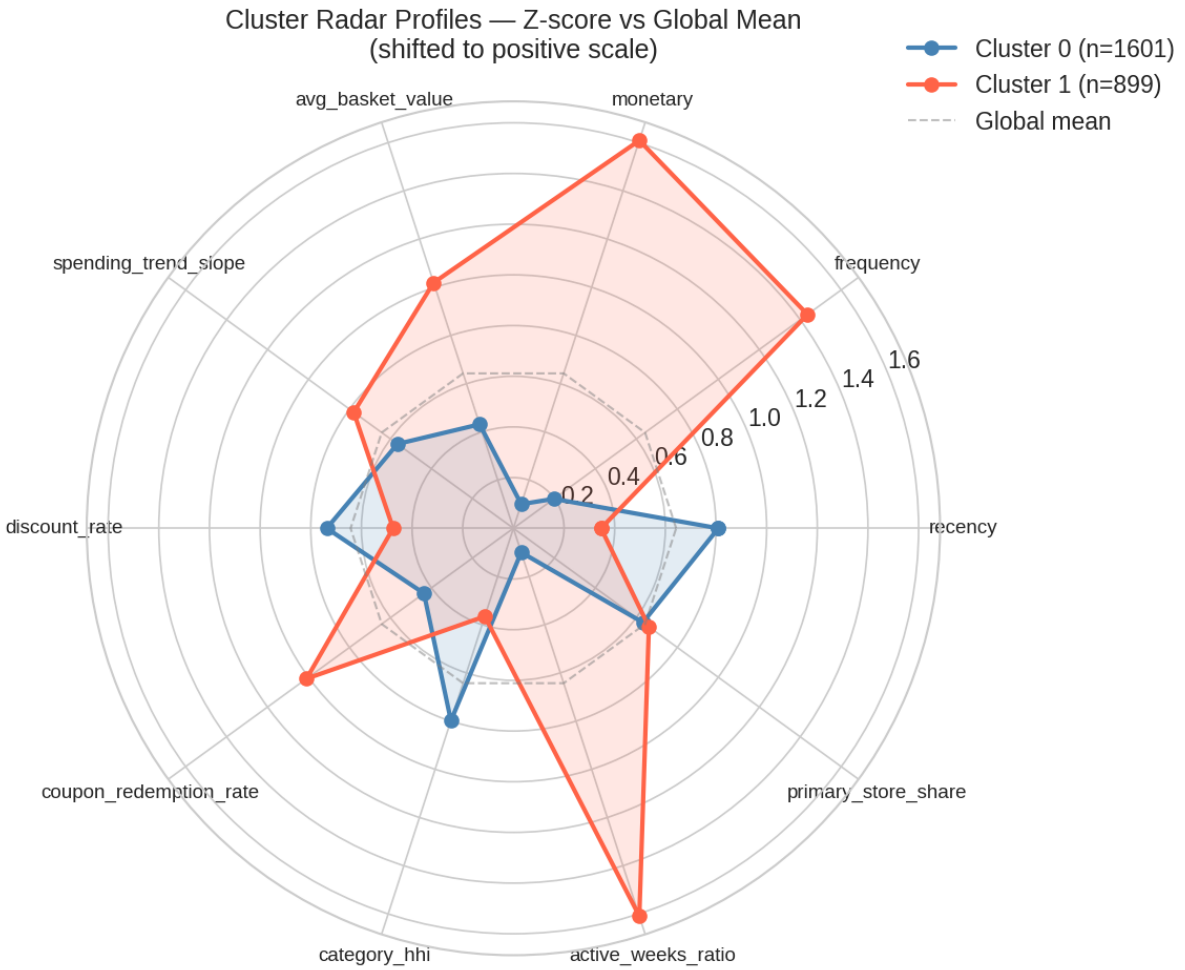

Figure 14: Radar chart comparing z-score profiles against population mean

8.3 Demographic Cross-Validation

Demographic data is available for 32% of households (801 of 2,500). As shown in Table 8, the middle-income bracket (35–74K) dominates both clusters. The key distinction is at the extremes: Cluster 1 has notably higher representation in upper-middle and high income brackets (23% and 10% vs. 16% and 3% in Cluster 0), and lower representation in the low-income bracket (24% vs. 32%). This weak income gradient is directionally consistent — higher-value shoppers tend toward higher income — but the overlap is substantial and the 32% coverage means these findings are indicative only, not conclusive.

Feature	**Cluster 0**	**Cluster 1**
recency	36.03	6.95
frequency	58.90	201.61
monetary	1,405.71	6,459.31
avg_basket_value	27.69	38.92
avg_basket_size	452.98	1,222.01
std_basket_value	25.35	34.86
n_unique_departments	10.48	15.47
category_hhi	0.362	0.319
national_brand_ratio	0.703	0.721
discount_rate	0.156	0.143
coupon_redemption_rate	0.001	0.016
campaign_exposure_count	1.02	6.20
active_weeks_ratio	0.350	0.728
spending_trend_slope	−0.003	0.098
primary_store_share	0.707	0.714
Cluster size	**1,601 (64%)**	**899 (36%)**

Table 7: Mean Feature Values per Cluster

Cluster	**Low (<35K)**	**Middle (35–74K)**	**Upper-middle (75–149K)**	**High (≥150K)**
Cluster 0	32.0%	48.0%	16.0%	3.0%
Cluster 1	24.0%	44.0%	23.0%	10.0%

Table 8: Income distribution by cluster

9 Business Recommendations

9.1 Answer to the Research Question

Yes, meaningful behavioral groups exist within the 2,500-household population. K-Means and Hierarchical clustering independently recover the same two-segment structure with an ARI of 0.552, confirming the split is present in the data and not an artifact of any single algorithm's assumptions. The segments differ substantially on monetary value (4.6× gap), shopping frequency (3.4× gap), active weeks ratio (2.1× gap), and campaign exposure (6.1× gap) — differences large enough to justify clearly differentiated marketing strategies. Treating the full base uniformly would mean applying high-investment retention tactics to 1,601 low-engagement households while potentially under-serving the 899 high-value core.

9.2 Segment Strategies

Cluster	Name	Size	Top Signal	Recommended Action	KPI to Track
0	Passive Occasionals	1,601 (64%)	Low frequency, low `active_weeks_ratio`, flat trend	Reactivation campaigns with low-cost incentives; frequency nudges via personalized weekly offers	Trip frequency, active weeks ratio
1	Committed Regulars	899 (36%)	High monetary, high frequency, positive trend slope	Loyalty reinforcement, category expansion cross-sell, premium tier enrollment	Monetary growth, category breadth, retention rate

Table 9: Segment Strategy Summary

Cluster 0 — Passive Occasionals: These 1,601 households currently contribute low total value and show no organic growth trend. The priority intervention is frequency stimulation: targeted weekly offer reminders personalized to their limited category history, low-barrier incentives (e.g., threshold-based discounts on categories they already purchase) designed to increase trip regularity without heavy margin investment. Success should be measured by change in `active_weeks_ratio` and trip frequency over the following

quarter. Given the low baseline, even modest frequency improvement across this large group produces material revenue upside.

Cluster 1 — Committed Regulars: These 899 households are the retailer's highest-priority retention asset, generating 4.6× the monetary value of Cluster 0 with a positive spending trajectory. The recommended strategy has two components: first, loyalty reinforcement through recognition programs (VIP tiers, early access to promotions) that reward and cement existing behavior without relying on discounts — their below-average discount rate confirms price is not the driver of their loyalty. Second, category expansion: they currently shop 15.5 departments on average with room to grow; targeted cross-sell into adjacent categories (e.g., a GROCERY-heavy household introduced to PRODUCE or NUTRITION) can increase basket breadth and HHI reduction. Success should be measured by retention rate, total monetary growth, and number of active departments per household.

10 Conclusion

10.1 Summary

This study identifies two distinct behavioral segments within the 2,500-household population: Passive Occasionals (64%, low engagement, flat trajectory) and Committed Regulars (36%, high value, growing spend). K-Means delivers the strongest geometric performance (Silhouette 0.206, CH 599) and moderate cross-algorithm stability (ARI 0.552 with Hierarchical). The most actionable insight is the 4.6× monetary gap between segments combined with the opposite spending trend directions, which together make a clear case for differentiated retention investment rather than uniform mass marketing. This study demonstrates that customer segmentation based on behavioral transaction data yields robust, interpretable household groups. The clustering pipeline — from raw transaction data to behavioral segment profiles — provides a replicable framework for any retailer with loyalty program data.

10.2 Limitations

- No ground truth: cluster validity is judged by interpretability and metric scores, not by comparison to known labels.
- Different feature sets or preprocessing choices would produce different segments — all potentially valid.
- Demographic coverage: approximately 68% of households lack demographic data, limiting post-hoc profile enrichment.
- Static snapshot: the 2-year window produces a fixed segmentation. In practice, households migrate between segments over time, requiring periodic model retraining.
- Metric limitations: Silhouette Score and Davies-Bouldin Index measure geometric cluster quality, not business relevance. A cluster with a high Silhouette score may still yield no actionable distinction.

10.3 Future Work

- Soft clustering with Gaussian Mixture Models: assign each household a probability of belonging to each segment, capturing uncertainty at cluster boundaries.
- Temporal segmentation: track how households transition between segments across rolling time windows to identify lifecycle patterns.
- Supervised integration: use cluster labels as features in a churn prediction model to assess whether segment membership predicts future behavior.

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3. Ester, M., Kriegel, H. P., Sander, J., & Xu, X. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise. In *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining (KDD-96)* (pp. 226-231).